

Lesson 9 Set-Theoretic Proofs

1. Prove the following proposition:
Suppose $a, b \in \mathbb{Z}$, $A = \{x \in \mathbb{Z} : a|x\}$ and $B = \{x \in \mathbb{Z} : b|x\}$, then $A \subseteq B$ if and only if $b|a$.
2. Let $A = \{x \in \mathbb{Z} : 6|x\}$ and $B = \{15n - 9m : n, m \in \mathbb{Z}\}$. Prove that $A \subseteq B$ but $A \neq B$.
3. Let A, B, C denote sets. Consider the following set relation.
$$(A \setminus B) \setminus C = A \setminus (B \cup C)$$
Use a Venn diagram illustration to show the above equivalence.
4. Let A and B be subsets of integers, and we define:
$$A + B = \{a + b : a \in A, b \in B\}$$
Prove that if A is the set of negative integers and B is the set of even integers, then $A + B = \mathbb{Z}$.
5. Let A and B be any subsets of integers and define:
$$AB = \{ab : a \in A, b \in B\}$$
Let A be the set of even integers. Prove that if $AB = A$, then $1 \in B$ or $-1 \in B$.
Hint: You may start with $2 \in AB$, explain why, and continue from there.

Let A, B, C be sets. Prove or disprove:

6. If $A \subseteq B$ then $A \cup B = B$.
7. If $A \setminus B = A$, then A and B are disjoint.
8. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.
9. If $A \subseteq C$ and $B \subseteq C$, then
 - a) $A \cap B \subseteq C$
 - b) $A \cup B \subseteq C$
10. If $A \setminus B = \emptyset$, then $B \setminus A = \emptyset$.
11. $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$
12. $B = \overline{A}$ if and only if A and B are disjoint.
13. If $A \setminus B = B \setminus A$, then $A = B$.
14. $A = B$ if and only if $A \cap B = A \cup B$.
15. $A \setminus B$ and $B \setminus A$ are disjoint.
16. $A \setminus B = \emptyset$ if and only if $A \subseteq B$.